Big Data Mining

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Lecture 2: Data Preparation

Spring 2026

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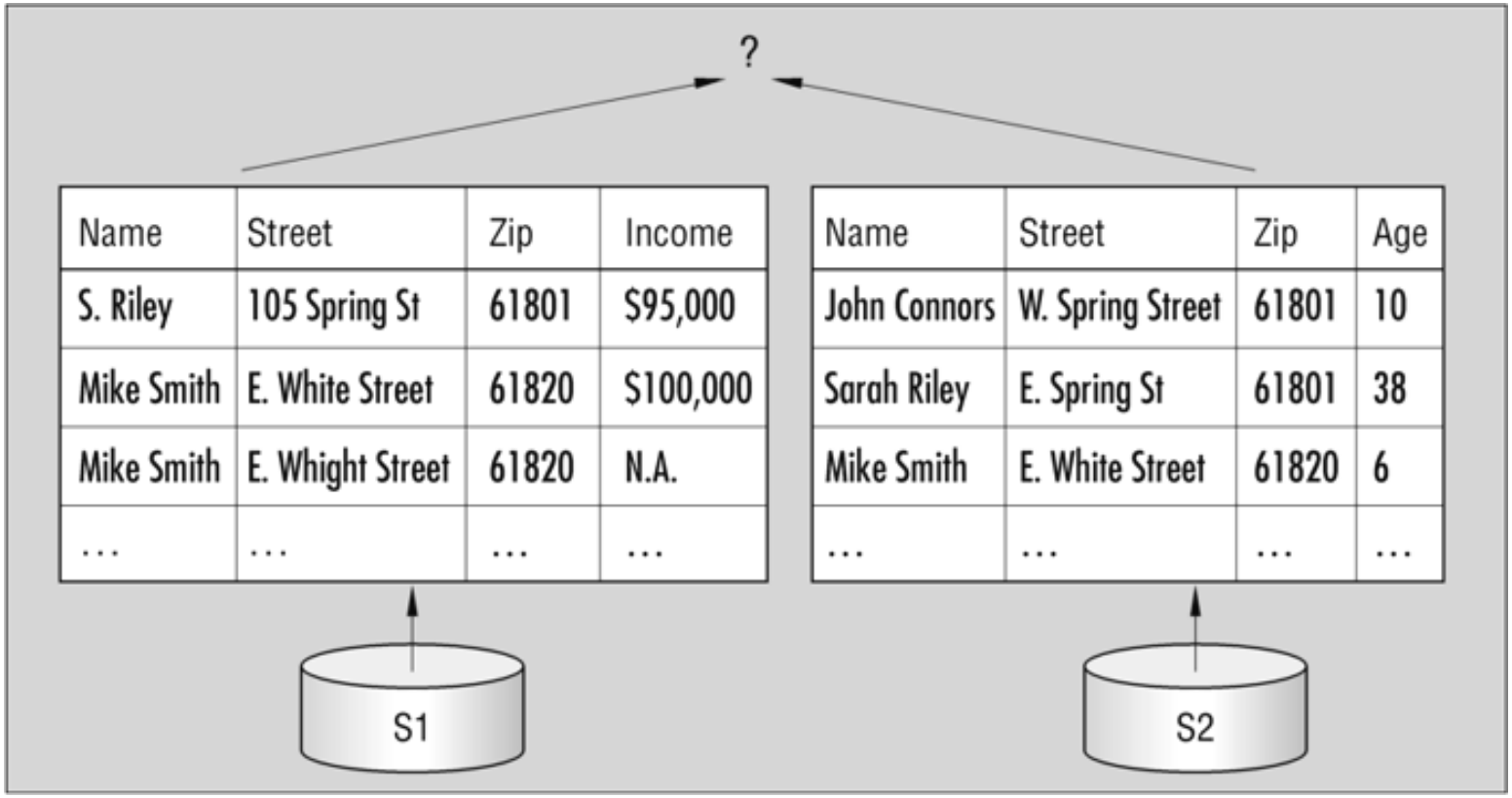
Main Sources

- Data Mining, The Textbook (2015) by Charu Aggarwal (Chapter 2) + slides by Lijun Zhang
- Introduction to Data Mining 2nd edition (2019) by Tan et al. (Chapter 2)
- Data Mining Concepts and Techniques, 3rd edition (2011) by Han et al. (Chapter 3)

Data integration

Data integration is not easy

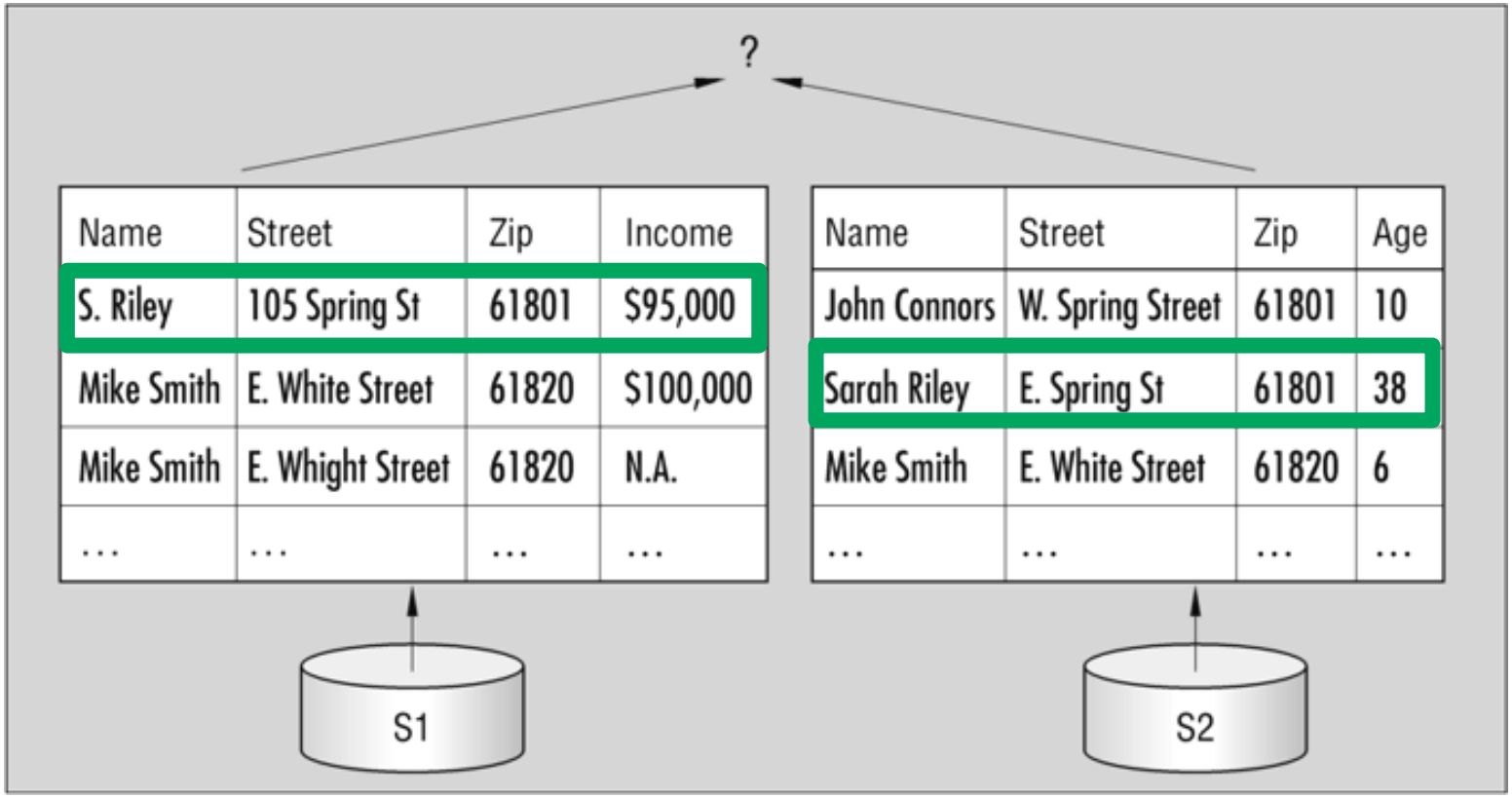


Name	Street	Zip	Income
S. Riley	105 Spring St	61801	$95,000
Mike Smith	E. White Street	61820	$100,000
Mike Smith	E. Whight Street	61820	N.A.
…	…	…	…

Name	Street	Zip	Age
John Connors	W. Spring Street	61801	10
Sarah Riley	E. Spring St	61801	38
Mike Smith	E. White Street	61820	6
…	…	…	…

Lu et al. 2013

Data integration is not easy



Lu et al. 2013

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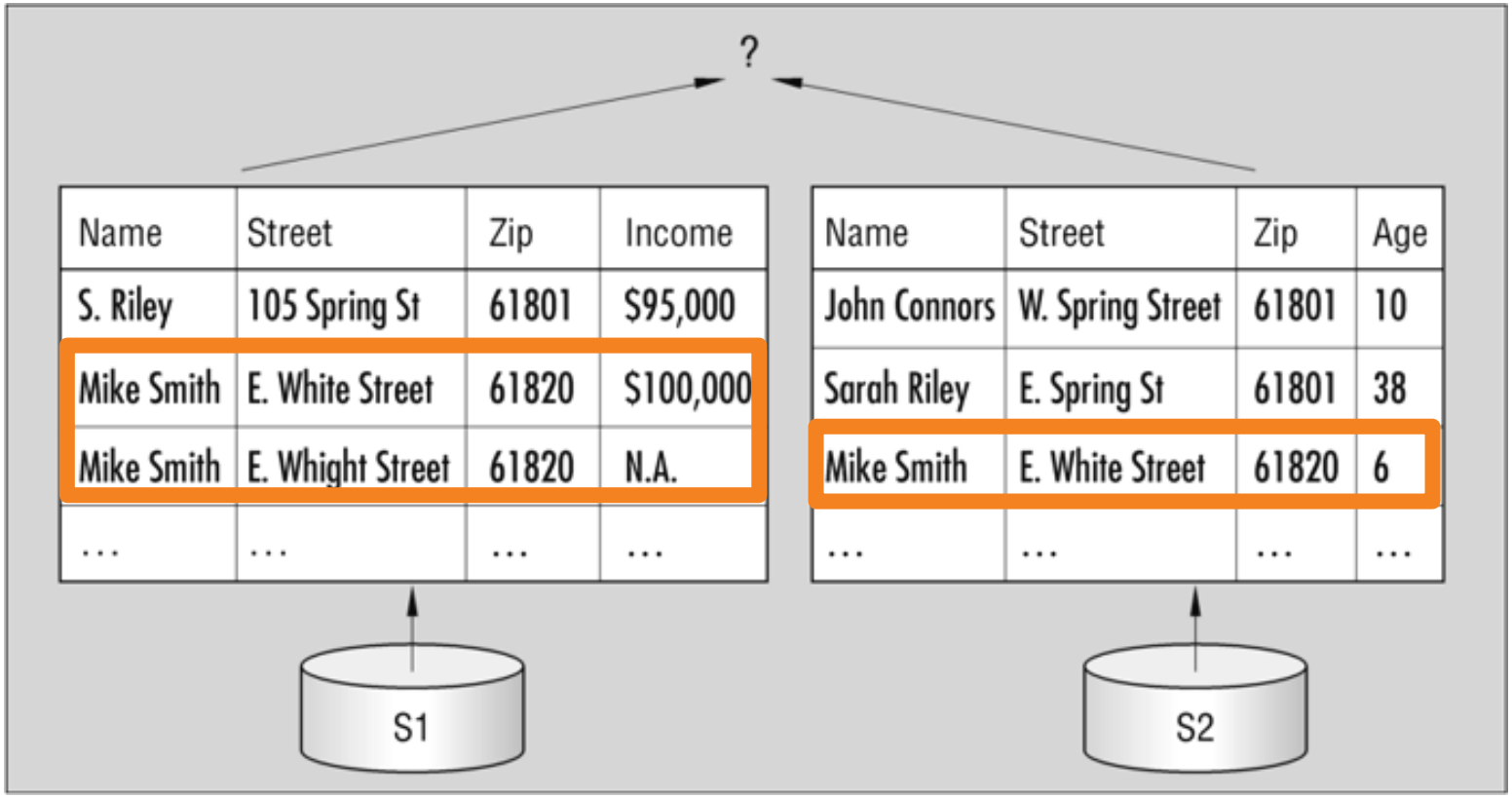


Lu et al. 2013

Data integration aspects

Remember: difference between DB schema and DB state

- **Schema integration**
  - Bring different schemata together
  - Equal concepts should be represented with equal types
- **Object matching** / Entity identification
  - Equal entities should be equally identified across datasets (unless re-identification forbidden by policy)

Data integration aspects (cont.)

- Redundancy analysis
  - Sometimes data needs to be integrated because different sets are row-incomplete
  - Sometimes those sets do not form a partition ⇒ there will be repeated entities to be removed
- Resolution of value conflicts
  - Same entity, different attribute values

Data cleaning

Why data cleaning?

- Data collection technologies are inaccurate
  - Sensors
  - Optical character recognition
  - Object recognition
  - Speech-to-text data
- Humans make mistakes when doing manual data entry
  - Entering numbers (e.g., accounting): 1% to 5% error rate
  - Even a 0.5% error rate, which is optimistic, can be problematic
- Software errors, data extraction/loading errors, and so on

What is data cleaning?

It is a process by which data records are

modified or **deleted**

until each record passes

data validity criteria ✅

Data validity criteria (1)

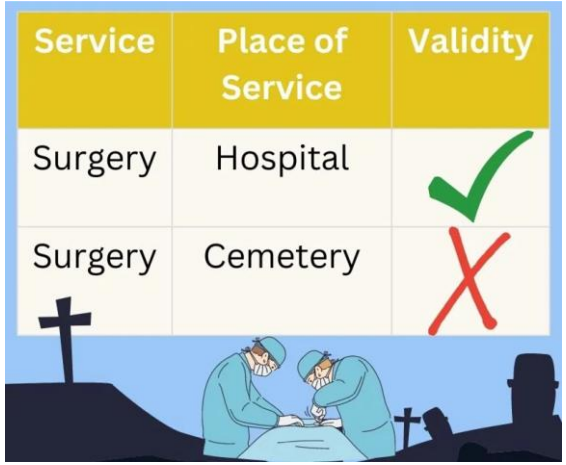
- **Mandatory** constraints: certain columns cannot be empty.
- **Data-type** constraints: values in a column must be of a particular datatype
- **Range** constraints: numbers or dates should fall within a certain range
- **Regular expression** patterns: e.g., phone numbers [0-9]{9}
- **Complex** constraints: e.g., IBAN, DNI numbers have specific formulas for some components

Data validity criteria (2)

- **Unique** constraints: a field, or a combination of fields, must be unique
- **Set-membership** constraints: values in a column come from a set of discrete values or codes
- **Foreign-key** constraints: set membership constraint where valid values in a column are defined in a column of another table that contains unique values

Data validity criteria (3)

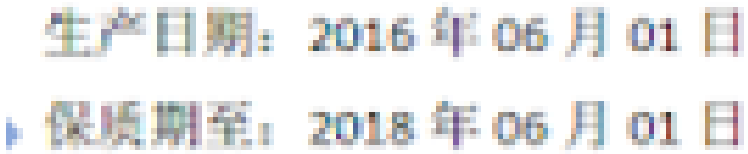
- **Cross-field validation**: certain conditions involving multiple fields must hold, e.g.:
  - percentages add up to 100
  - discounted price ≤ regular price
  - …

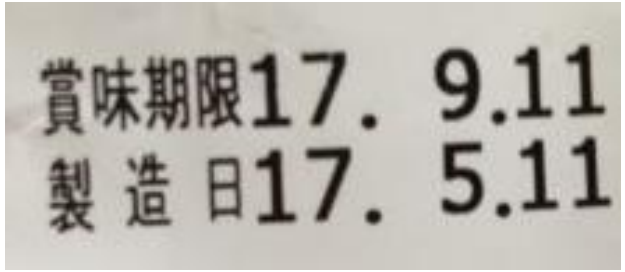


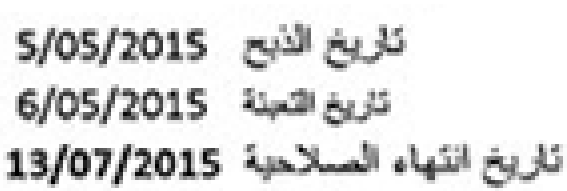
[source]

Data validity examples

You see this in a package … how do you decide whether the product is expired or not?







Handling missing entries

Why is a value missing?

- **Missing Completely at Random (MCAR)**
  - Missingness of a value is independent of observable attributes
- **Missing at Random (MAR)**
  - Missingness has statistical dependencies with an observable attribute
  - We can fill in values based on other attributes, but this is likely to introduce a bias in the analysis
- **Missing Not at Random (MNAR)**
  - Missingness depends deterministically on an observable attribute
  - In general this is informative, non-ignorable missingness

In general, it is **not** possible to know which one is the case just by looking at the data

Handling missing entries: options

- **Delete** the data record containing missing entries
- Estimate or **Impute** the Missing Values
  - Additional errors may be introduced
  - Good under certain conditions (e.g., Matrix Completion)
- Some algorithms can work with missing data

Exercise

Handling missing data (**write WHY**)

- 5% of student records at a university have no “civil status” (single, married, ...)
  Drop records? Impute value, how? Write your reasons
- 5% of smokers in a study of the effects of tobacco on health had no year of birth
  Drop records? Impute value, how? Write your reasons
- 5% of records of sales of a company have zip code but no province
  Drop records? Impute value, how? Write your reasons
- Temperature sensor at weather station failed at random intervals during one day, total downtime 6 hours, max continuous downtime 15 minutes
  Drop records? Impute value, how? Write your reasons
- Same sensor failed during one night, downtime 6 hours continuous
  Drop records? Impute value, how? Write your reasons

Handling Incorrect and Inconsistent Entries

- Inconsistency detection
  - E.g., full name and abbreviation do not match
- Domain knowledge
  - Human age cannot reach to 800 years (yet?)
- Data-centric methods
  - Outlier detection

Scale

- Features have different scales
  - Age might in 10s, Salary in 1000s
- Euclidean distance between vectors with components in different scales makes no sense

Normalization

- Standardization ("z-scoring")
  - Ensures each column has mean 0 and stdev 1

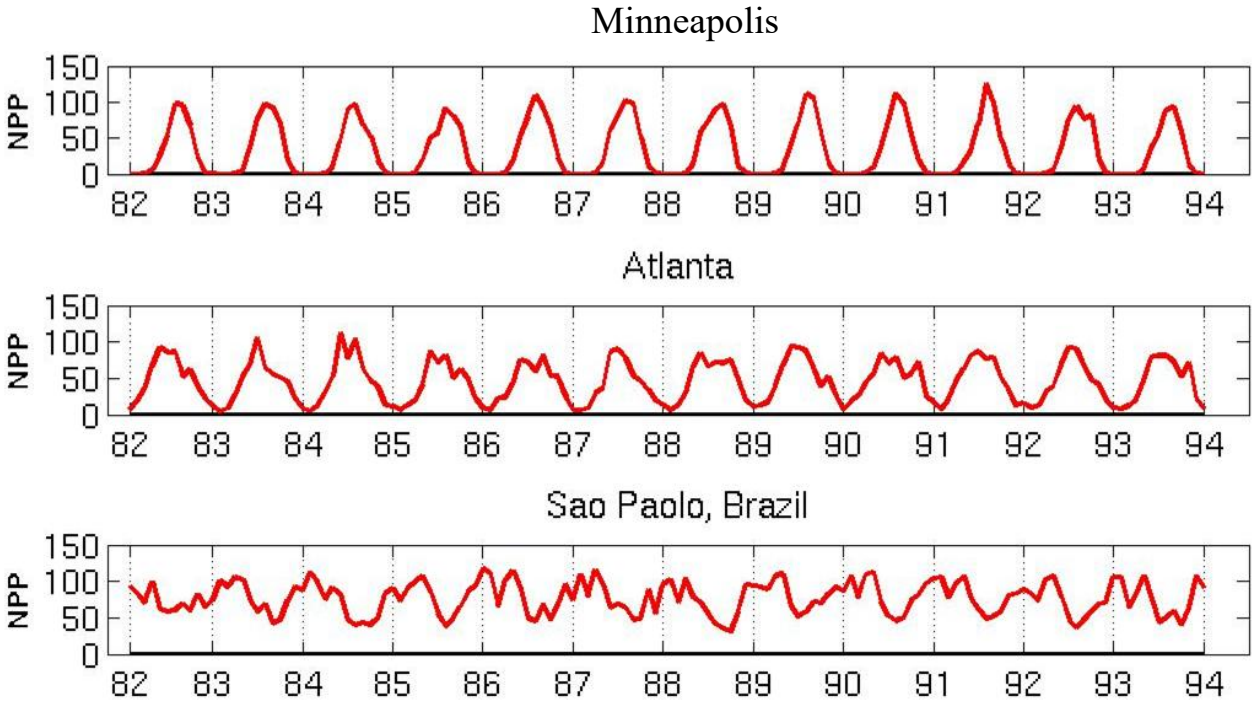
$$z_i = \frac{x_i - \mu}{\sigma}$$

- Min-Max Scaling
  - Ensures each column is in [0,1]

$$z_i = \frac{x_i - \min}{\max - \min}$$

Both are quite sensitive to outliers

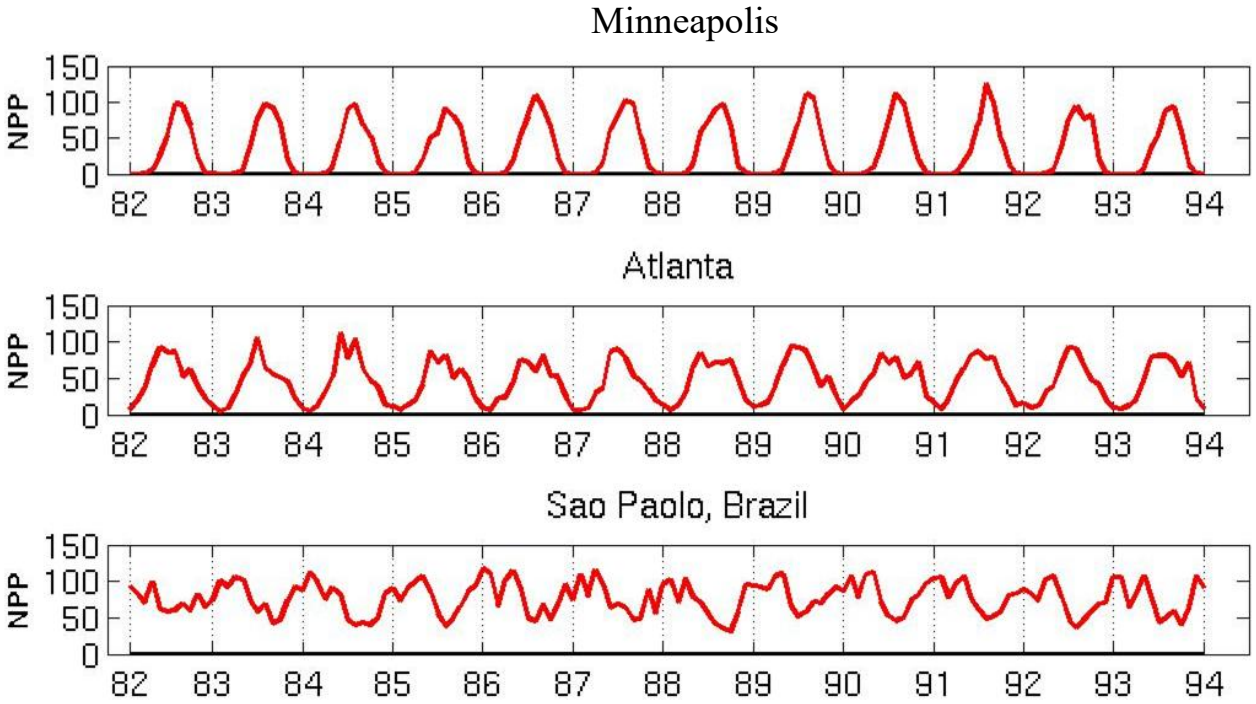
Example: seasonal standardization



Net Primary Production (NPP) is a measure of plant growth used by ecosystem scientists.

	Minneapolis	Atlanta	Sao Paolo
Minneapolis	1.0000	0.7591	-0.7581
Atlanta	0.7591	1.0000	-0.5739
Sao Paolo	-0.7581	-0.5739	1.0000

Example: seasonal standardization



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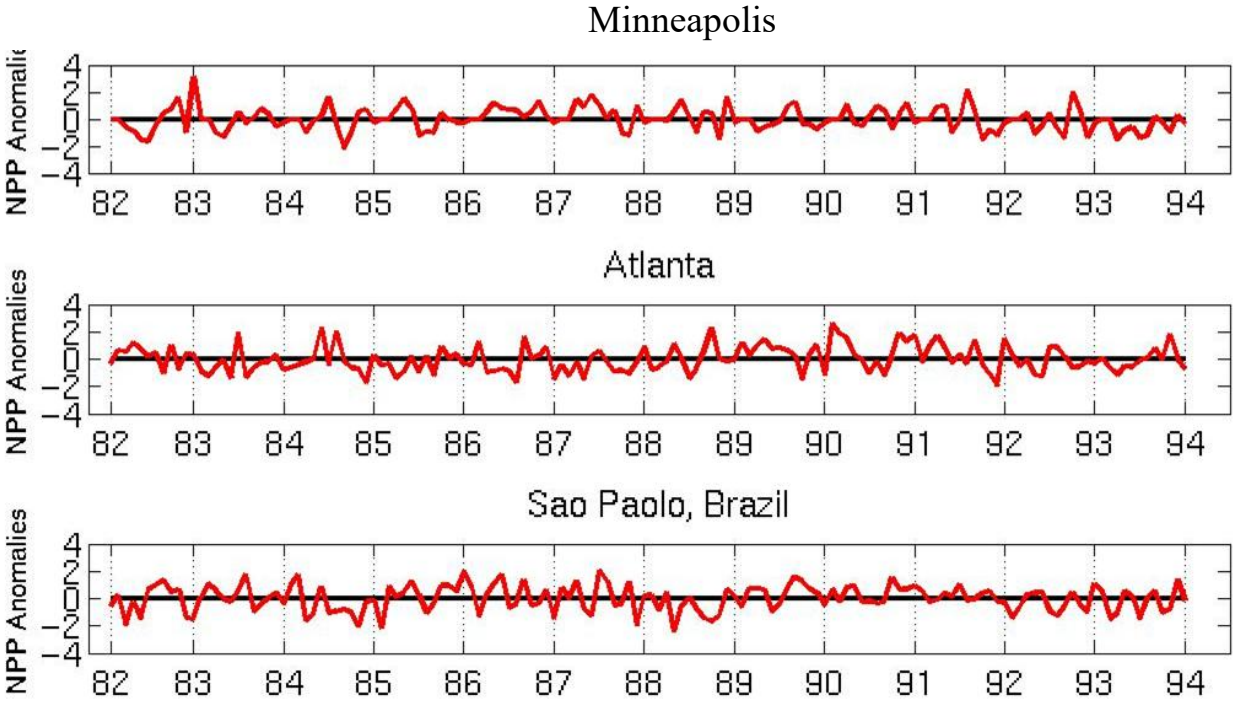
Spurious correlations between time series

Introduction to Data Mining 2nd edition (2019) by Tan et al. (Chapter 2)

Example: seasonal standardization



Normalized using monthly Z Score:

Subtract off monthly mean and divide by monthly standard deviation

	Minneapolis	Atlanta	Sao Paolo
Minneapolis	1.0000	0.0492	0.0906
Atlanta	0.0492	1.0000	-0.0154
Sao Paolo	0.0906	-0.0154	1.0000

Adjusted correlations between time series

Summary

Things to remember

- What does it mean to do data cleaning
  - Modify or delete records from a database until the database satisfies a set of validity criteria
- What to do with missing values
  - It depends on **why** those values are missing

Related Exercises

- Exercises 3.7 of Data Mining Concepts and Techniques, 3rd edition (2011) by Han et al.
- Exercises 2.6 of Introduction to Data Mining, Second Edition (2019) by Tan et al.
  - Mostly the first exercises, say 1-6

Data Reduction and Transformation

Data reduction and transformation

- Sampling
  - $\simeq$ “Less rows”
- Dimensionality Reduction or Feature Selection
  - $\simeq$ “Less columns”

Why reduce/transform data?

- Advantages
  - Reduce space complexity
  - Reduce time complexity
  - Reduce noise
  - Reveal hidden structures (e.g., manifold learning)
- Disadvantages
  - Information loss

Sampling for static data

- **Uniform** random sampling
  - with/without replacement
- **Biased** sampling
  - e.g., emphasize recent items
- **Stratified** sampling
  - Partition data in strata, sample in each stratum

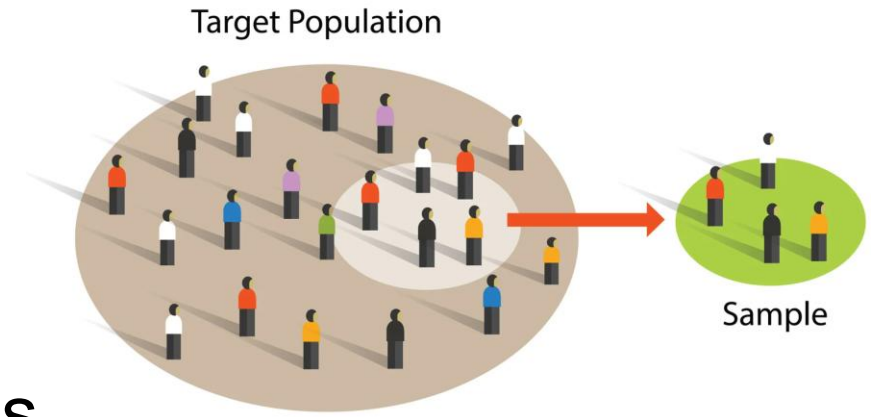


Image source

Sampling example

- Suppose there is a population of 10,000 people that contains 100 millionaires
- Uniform random sample of 100 people
  - In expectation, one millionaire will be sampled
  - There is $\simeq$ 37% chance no millionaires are sampled: $(1 - 0.01)^{100}$
- **Stratified Sampling** of 100 people
  - Uniform Random Sampling of 1 from 100 millionaires
  - Uniform Random Sampling of 99 from non-millionaires

Sampling from data streams

- Suppose you want to give away for free 10 VIP passes at a concert
  - You want everybody to have exactly the same chance of getting the VIP pass, independently on when they arrived, as long they arrive before the concert starts
  - Once people leave the entrance area they become impossible to find, so if you win, you should receive the VIP pass at the door
  - People arrive in sequence, and you do not know how many people will arrive
- **Reservoir sampling algorithm** … seen in the stream processing part

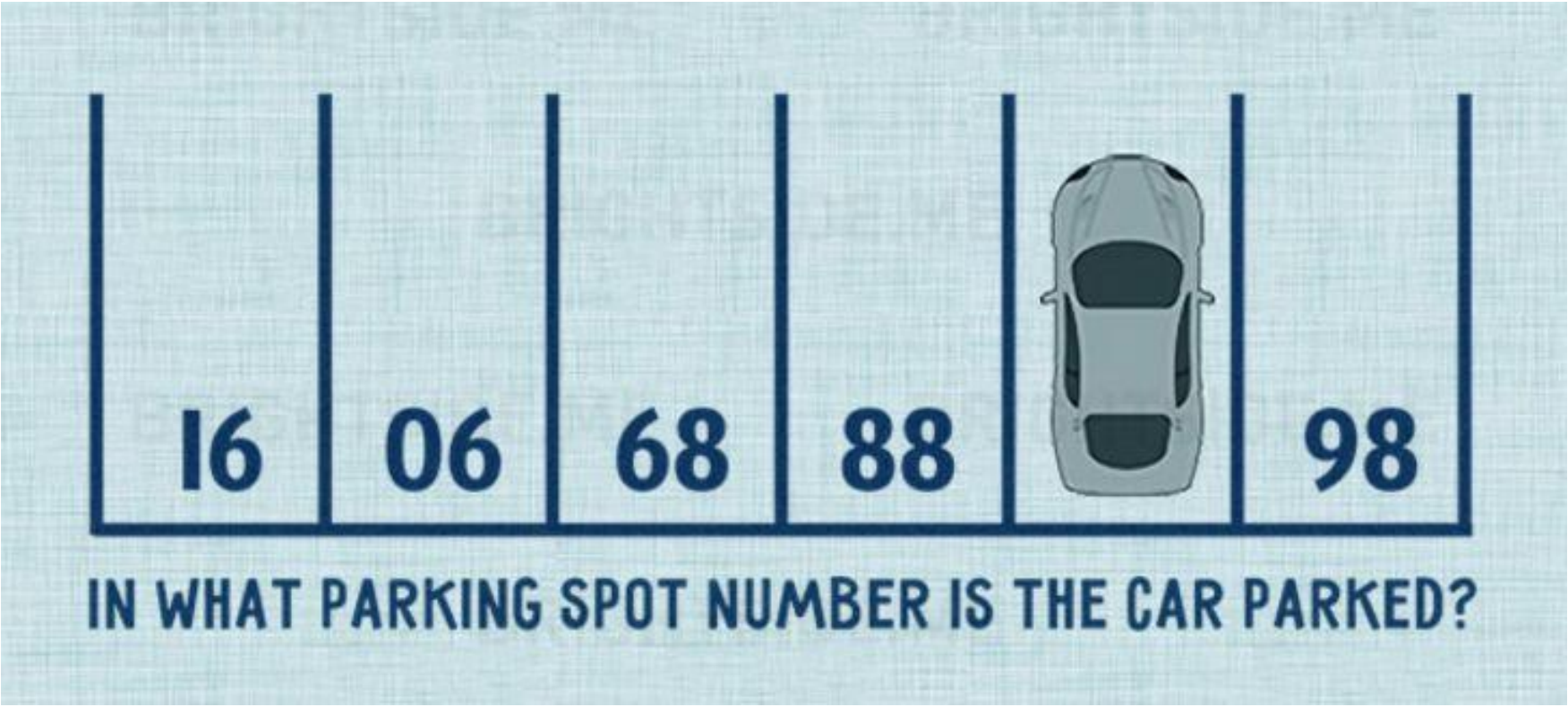
Reducing data dimensionality

Note: PCA/SVD covered well in other courses, won’t be part of our exam

Feature selection

- **Unsupervised** Feature Selection
  - Using the performance of unsupervised learning (e.g, clustering) to guide the selection
- **Supervised** Feature Selection
  - Using the performance of supervised learning (e.g., classification) to guide the selection

Remember: rotating an axis may help



Source: [Centauro Blog (2017)]

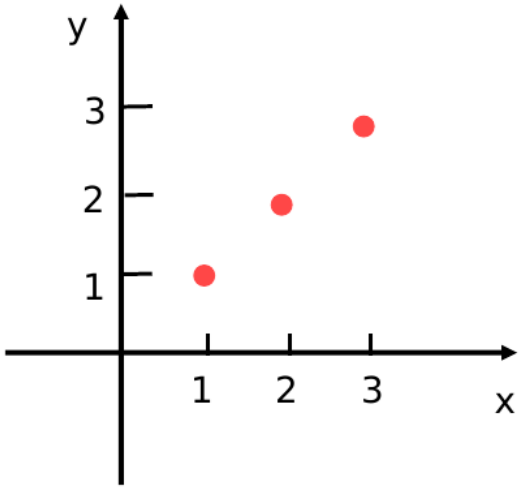
Dimensionality reduction with axis rotation (perfect case)

- Motivation: three points in a line in two-dimensional space

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$\mathbf{x}_3 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$



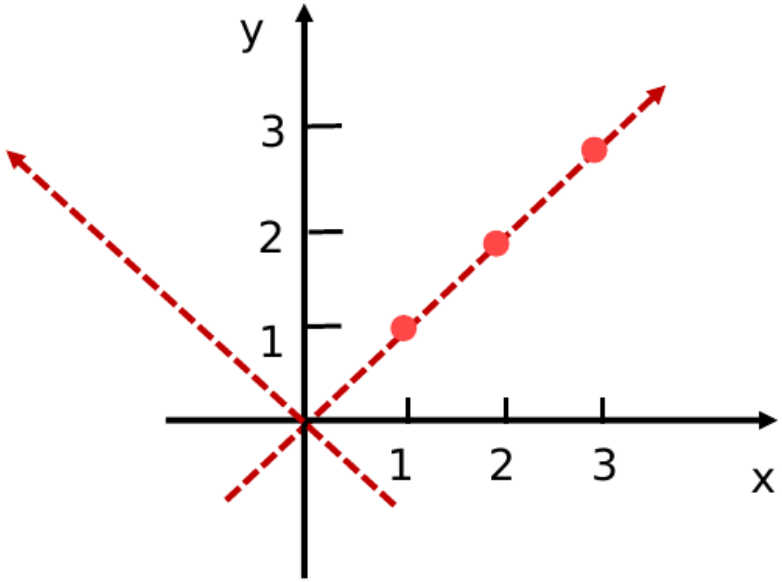
Dimensionality reduction with axis rotation (perfect case, cont.)

- Coordinates after axes rotation

$$\mathbf{x}_1 = \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix}$$

$$\mathbf{x}_3 = \begin{bmatrix} 3\sqrt{2} \\ 0 \end{bmatrix}$$



Dimensionality reduction with axis rotation (perfect case, cont.)

- Coordinates after axes rotation

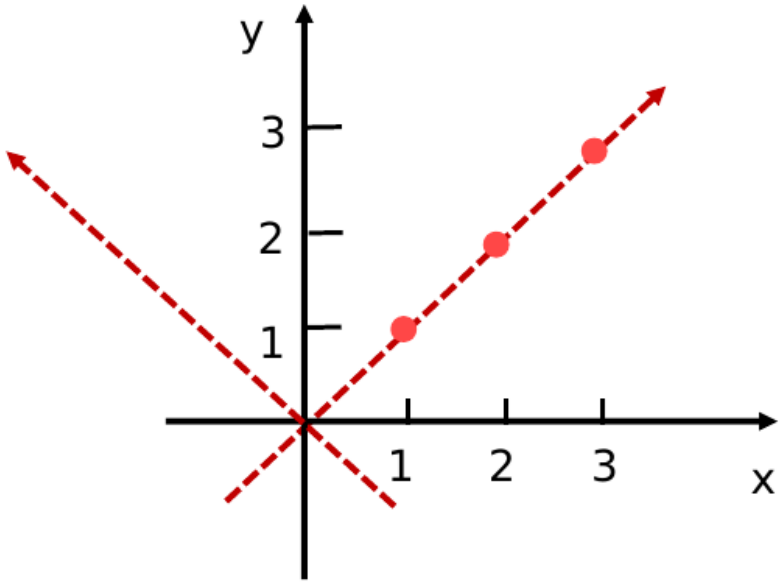
Drop second coordinate, **no** information is lost.

2D data reduced to 1D data

$$\mathbf{x}_1 = \begin{bmatrix} \sqrt{2} \\ \cancel{0} \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} 2\sqrt{2} \\ \cancel{0} \end{bmatrix}$$

$$\mathbf{x}_3 = \begin{bmatrix} 3\sqrt{2} \\ \cancel{0} \end{bmatrix}$$



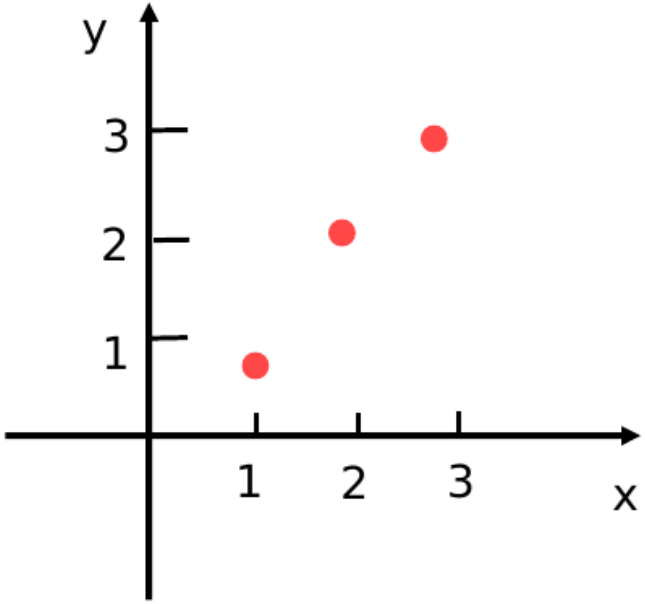
Dimensionality reduction with axis rotation (noisy case)

- Suppose points don't lie exactly on a line

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0.9 \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} 2.1 \\ 2 \end{bmatrix}$$

$$\mathbf{x}_3 = \begin{bmatrix} 2.9 \\ 3.1 \end{bmatrix}$$



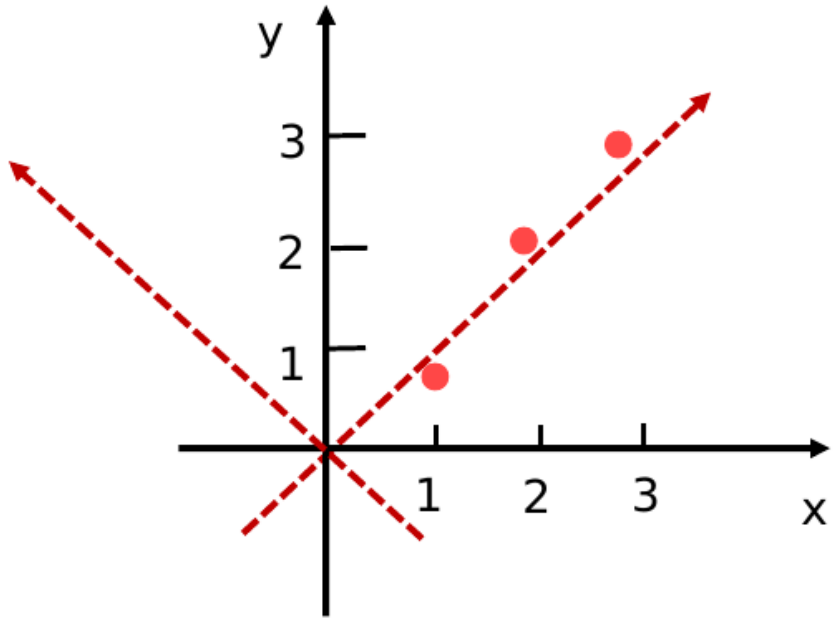
Dimensionality reduction with axis rotation (noisy case, cont.)

- Suppose points don't lie exactly on a line

$$\mathbf{x}_1 = \begin{bmatrix} 1.34 \\ 0.07 \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} 2.89 \\ 0.07 \end{bmatrix}$$

$$\mathbf{x}_3 = \begin{bmatrix} 4.24 \\ -0.14 \end{bmatrix}$$



Dimensionality reduction with axis rotation (noisy case, cont.)

- Suppose points don't lie exactly on a line

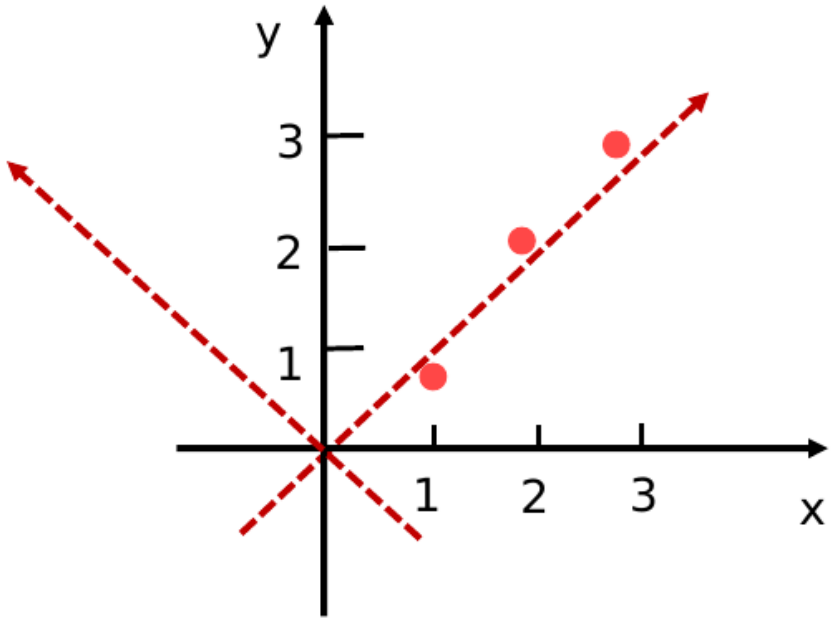
Drop second coordinate, **some** information is lost.

2D data reduced to 1D data

$$\mathbf{x}_1 = \begin{bmatrix} 1.34 \\ \cancel{0.07} \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} 2.89 \\ \cancel{0.07} \end{bmatrix}$$

$$\mathbf{x}_3 = \begin{bmatrix} 4.24 \\ \cancel{-0.14} \end{bmatrix}$$



How does this work in reality?

- Change of axes removes correlated features and reduces dimensionality
- Techniques
  - Principal Component Analysis (PCA)
  - Singular-Value Decomposition (SVD)
  -

Summary

Things to remember

- Data sampling methods
- Why would we want to reduce dimensionality?
- What are the main techniques for doing so

Related Exercises

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